

Section 6 — N(Z): The Normality-Defect Index ("ZEUS")

Status: HYPOTHESIZED tier. Not asserted as isomorphic to any Pellis construct. Proposed as a single bounded dial that gives the existing eigenvector-condition-number / Henrici departure-from-normality discussion (§4) a normalized index, with the rotation (normal-operator) regime and the exceptional-point (defective-operator) regime as its two literal poles.

6.1 Motivation

§4 established that the leverage operator J moves between two qualitatively distinct regimes:

- **Normal regime:** J normal ($JJ^* = J^*J$) — e.g. the pure-rotation case $R(\theta) = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$. Eigenvalues $e^{\pm i\theta}$; eigenvectors $(1, \mp i)/\sqrt{2}$ — fixed, orthogonal, independent of θ . Eigenvector condition number $\kappa(V) = 1$.
- **Defective limit:** J at the exceptional point ($\gamma_1\gamma_2 = -(\epsilon - \delta)^2$) — eigenvalues coalesce ($\lambda_+ = \lambda_-$), eigenvectors collapse parallel, geometric multiplicity drops to 1, $\kappa(V) \rightarrow \infty$.

These are not two arbitrary reference points — they are the two structural poles of operator theory itself (normal vs. maximally non-normal). Naming them as fixed anchors of a bounded index is a definitional move, not a claim that they converge.

6.2 Definition

Let $\Delta(J)$ be the Henrici departure-from-normality:

$$\Delta(J) = \sqrt{\|J\|_F^2 - \sum_i |\lambda_i|^2} \geq 0$$

$\Delta(J) = 0$ exactly when J is normal. Define the bounded, signed N(Z) index:

$$N(Z) = 1 - \frac{2\Delta(J)}{\Delta(J) + \|J\|_F} \in (-1, 1]$$

Pole	Operator state	Eigenvector geometry	$\kappa(V)$	$N(Z)$
$N = +1$	Normal (rotation regime)	Orthogonal, angle-invariant	1	+1
$N \rightarrow -1$	Maximally defective (exceptional point)	Parallel / coalesced	$\rightarrow \infty$	$\rightarrow -1$

6.3 "ZEUS" — naming convention for the index

Zipper Energy United Spectrum (ZEUS) names $N(Z)$ as an object, with each term keyed to a specific feature of the operator, not used as independent evidence of a mechanism:

Term	Referent	Basis
United Spectrum	$\lambda_+ = \lambda_-$ at $N \rightarrow -1$ — eigenvalues literally coalesce	Exact (derived in §3)
Energy	$\text{Re}(\lambda)$, dissipation/decay term carried from the REAL mapping (§5)	Hypothesis-tier, inherited
Zipper	Eigenvector state: orthogonal/distinct at $N = +1$ ("open"), parallel/merged at $N \rightarrow -1$ ("closed")	Interpretive label for a derived mechanism — naming convention, not independent proof

Note the asymmetry: "United Spectrum" names the $N = -1$ pole precisely; it does not separately name the $N = +1$ pole (maximally *split*, orthogonal spectrum). The name foregrounds the collapse end of the dial. This is consistent with naming conventions elsewhere in canon (SIGNET, SCALULAR) and does not affect the validity of $N(Z)$ itself — it is a property of the label, not the index.

6.4 Relationship to existing constructs

$N(Z)$ is **not** proposed as a replacement for $\kappa(V)$ or $\Delta(J)$ — both remain the primary, unbounded diagnostic quantities (§4). $N(Z)$ is a normalized presentation layer for communicating where a given instrument's operator sits on the normal $\leftrightarrow$ defective spectrum, bounded for comparison across instruments with different scales of $\|J\|_F$.

Open items (pending Pellis):

- Whether $\Delta(J)$ (vs. $\kappa(V)$ directly) is the correct quantity to normalize for this purpose.

2. Whether $N(Z) = -1$ should be reserved strictly for the exceptional point, or extended to the broader defective neighborhood around it.
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Drafted 2026-06-21 for insertion into the ERES-Pellis spectral bridge document, following §5 (Proposed Mapping). HYPOTHESIZED tier throughout, consistent with document-wide status convention.

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Credits

- **Joseph Allen Sprute (JAS)** — ZEUS naming (Zipper Energy United Spectrum), $N=+1/N=-1$ dial framing, conceptual direction.
- **Claude (Sonnet 4.6, Anthropic)** — formalization, Henrici departure-from-normality derivation, comparison table.
- **Stergios Pellis** — originating operator-theoretic framework (non-self-adjoint Jacobian, exceptional-point structure) this section extends. No claim of endorsement.

References

1. Henrici, P. (1962). Bounds for iterates, inverses, spectral variation and fields of values of non-normal matrices. *Numerische Mathematik*, 4(1), 24-40.
2. ERES-Pellis spectral bridge document, §§3-5 (toy model and exceptional-point derivation this section extends), 2026-06-20.